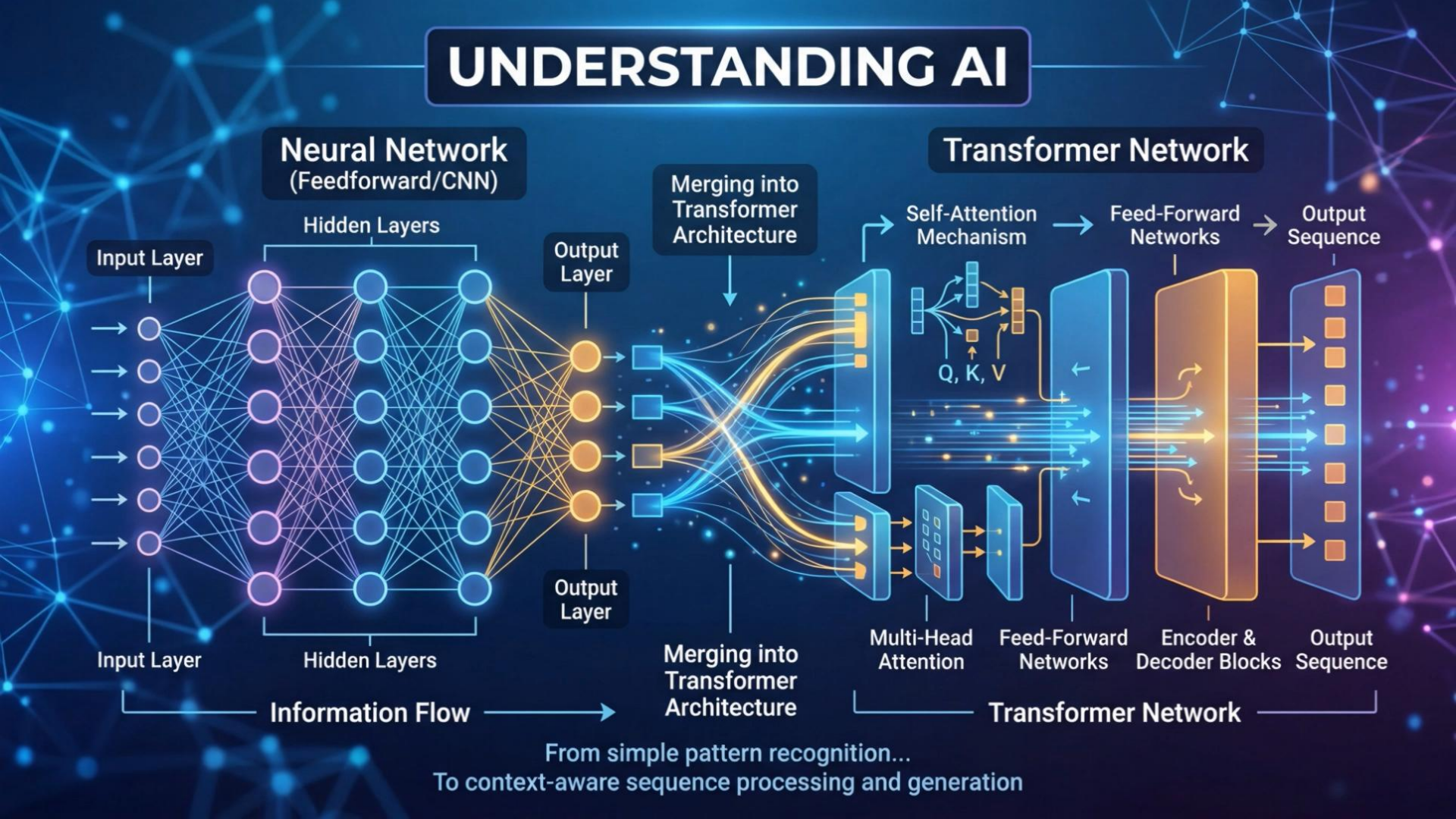
UNDERSTANDING AI



What We'll Cover Today

1

What Is AI?

Neural networks, key terminology, costs, shortcomings & bias

2

The cost of AI

Monetary cost of implementing an AI Data center.

3

Concerns About AI

Learning, cost, youth anxiety, cultural representation

4

Using AI for Research

How to ask good questions and avoid biased answers

5

AI Resources

Where can you find useful ideas.

huggingface.co

6

Fun Stuff

What anyone can do with AI.

PART 1

What Is AI?

Clearing the fog around a big buzzword

AI Is a Broad Category

THE FOUNDATION SPECTRUM

Simple Rules	Machine Learning	Neural Networks	Deep Learning
If/then logic, subsumption architecture, basic sorting	Learns patterns from data	Brain-inspired computing	Many layered neural nets

MODERN AI APPLICATIONS (all built on deep learning)

Large Language Models (LLMs)	Generators & Denoisers	Vision & Audio Recognition
ChatGPT, Claude, Gemini — trained on vast text to generate, summarize, and answer	Image generators (DALL·E, Midjourney), music AI — learn to create from noise	Facial recognition, medical imaging, speech-to-text, sound classification

Not everyone agrees on what deserves the label "AI." The Turing Test was proposed in 1950s. Our current LLMs have passed this test. But do they understand?

Neural networks date back to the 1940s — what changed is our computing power and data.

Key Terminology — Don't Let These Scare You

Neural Network

A system of interconnected mathematical "neurons" that process information in layers. Each layer finds patterns in the data, passing its findings to the next layer. Think of it like a very intricate game of telephone.

Tensors

Just a fancy word for organized lists of numbers. A single number is 0-dimensional. A list is 1D. A spreadsheet is 2D. A tensor can be many dimensions. AI models store everything — images, text, sounds — as tensors.

Machine Learning

The process where a computer improves its performance by learning from examples, rather than being given explicit rules. Show it 10,000 cat photos labelled "cat" and it learns to recognize cats.

Reinforcement Learning

Training through reward and consequence — much like training a dog. The AI takes an action, gets a "reward" signal if the outcome was good, and gradually learns which actions lead to better outcomes.

How Neural Networks Actually Learn

Think of it like teaching a child to recognize dogs:

1. **Show many examples**

   Thousands of labeled photos — "this is a dog," "this is not a dog"

2. **Get it wrong — adjust**

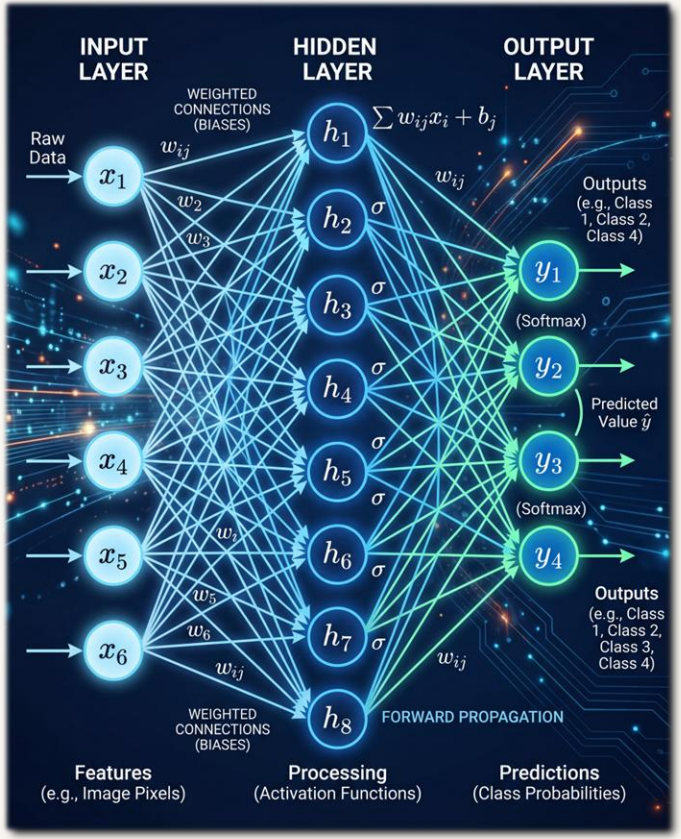
   Each mistake slightly updates how the network processes images

3. **Get it right — reinforce**

   Correct answers strengthen those pathways, wrong ones weaken them

4. **Repeat — millions of times**

   After enough examples, patterns emerge and generalize to new images



So — Do We Know How These Actually Work?

Yes. But it takes enormous effort to find out the details.

Large Language Models (LLMs)

ChatGPT, Claude, Gemini

Generators & Denoisers

DALL·E, Midjourney, Suno

Vision & Audio Recognition

Facial recognition, speech-to-text

Researchers can probe these models and map out what they have learned — which concepts cluster together, which patterns trigger which responses, even which individual neurons specialise in particular ideas.

But teasing out the specifics is massively complex work. These models contain billions of parameters, and understanding exactly why a model said what it said requires painstaking analysis — not a simple readout.

3Blue1Brown

PART 2

The cost of AI

Monetary cost of implementing an AI Data center.

The Cost of Teaching a Neural Network

Training large AI models is extraordinarily expensive — in dollars, energy, water and time

~$4M

Cost to train GPT-3

(2020)

$100M+

Estimated GPT-4 training cost

(2023)

Months

Time for a single training run

on thousands of GPUs

Once

How often we can afford to retrain from scratch

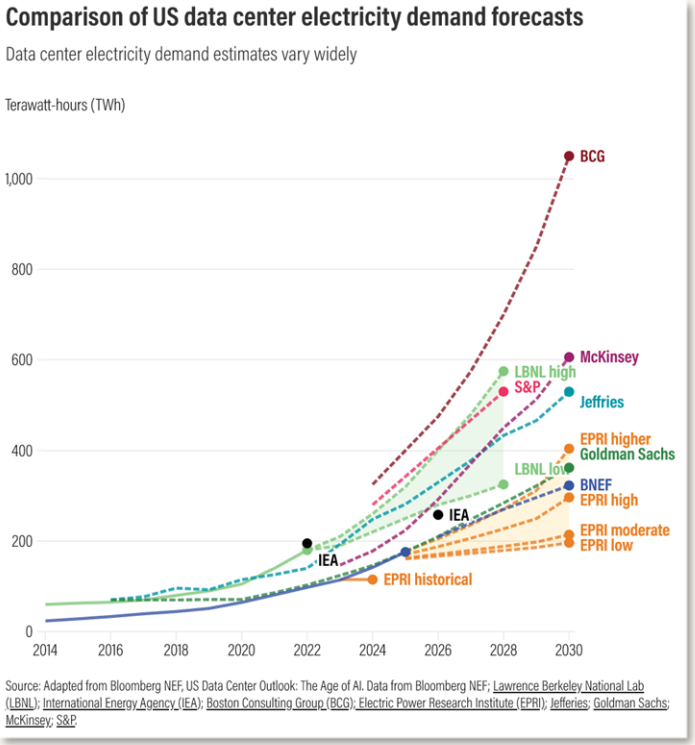
Existing models get updated, not rebuilt

Data center Costs: Water

Source: Google. Data center in The Dalles, Oregon releases water vapor and steam from cooling towers

Data center Costs: Power

- AI is part of our future.
- But in what form?
- But the competition is stiff and there can be many losers in this!



Source: World Resource Institute.

- Who pays to build the new power plants?
- If the AI market crashes, who carries the bill for the power plants?

Data center Costs: Consumer Electronics

AI Data centers need the fastest memory available. The demand is extremely high!
We as consumers would never need these high-speed memory in our electronics.
But the same factories that make the fast memory also make the consumer memory. They make more money by making the high-speed memory chips! This applies to other components as well!

Scalpers are seeing the trend and buying up memory and selling it at premium prices.

Scammers are also helping themselves, by offering too good to be true deals!

The pricing future is as uncertain as the AI market is!

G.SKILL 8GB DDR4 3200 (PC4 25600) Desktop Memory Model F4-3200C16S-8GVKB

- CAS Latency: CL38
- Voltage: 1.35V
- Timing: 16-18-18-38
- Recommend Use: AMD X570 / X470 / B550 / B450 Intel Z790 DDR4 / Z690 DDR4 / Z590 / Z490 / Z390 / Z370 / Z270 / Z170
- Model #: F4-3200C16S-8GVKB

~~$76.99~~
$69.99
Save: $7.00 (9%)
More options from $25.84 - $99.20
FREE SHIPPING from United States
ADD TO CART
Compare

Total Micro 8GB DDR4 SDRAM Memory Module - For Desktop PC, Workstation - 8 GB (1 x 8GB) - DDR4-3200/PC4-25600 DDR4 SDRAM - 3200 MHz Single-rank Memory - CL22 - 1.20 V - Non-ECC - Unbuffered ...

- Brand: Total Micro
- Capacity: 8GB
- Speed: DDR4 3200 (PC4 25600)
- Part Number: AB120718-TM
- Model #: AB120718-TM

$82.99
More options from $80.56 - $83.74
FREE SHIPPING from United States
ADD TO CART
Compare

G.SKILL Flare X5 8GB 288-Pin PC RAM DDR5 6000 (PC5 48000) Desktop Memory Model F5-6000J3038F8GH1-FX5

- CAS Latency: CL30
- Voltage: 1.35V
- Timing: 30-38-38-96
- Brand: G.SKILL
- Model #: F5-6000J3038F8GH1-FX5

~~$142.99~~
$129.99
Save: $13.00 (9%)
FREE SHIPPING from United States
ADD TO CART
Compare

(64)

Patriot Viper Elite 5 16GB 288-Pin PC RAM DDR5 6000 CL30(PC5 48000) Hynix Desktop Memory Model VEB516G6030W Compatible with AMD / Intel Gen14 Only

- CAS Latency: CL30
- Voltage: 1.35V
- Timing: 30-40-40-76
- Brand: Patriot
- Model #: VEB516G6030W

~~$219.99~~
$199.99
Save: $20.00 (9%)
More options from $46.99 - $226.32
FREE SHIPPING from United States
ADD TO CART
Compare

PART 3

Concerns About AI

Five things we should all be thinking carefully about

Why Aren't Neural Networks Doing Everything?

Hallucinations

AI confidently states false information. It doesn't know what it doesn't know — it predicts the most plausible-sounding next word, not the truest one.

Lack of Common Sense

AI has no lived experience, no embodied understanding. It cannot reason about physical reality or social context the way humans do.
Humans can also have false common sense but ai has no heuristics.

Context Limitations

Most AI models can only "remember" a limited window of a conversation. Long, complex discussions lose critical context.

AI Company CEOs believe bigger models will solve this. Researchers disagree!

Fragility

Tiny changes in wording can produce wildly different answers. The same question phrased differently may get opposite responses.

Energy Demands

Running AI inference (asking it questions) at scale consumes enormous power. Practical limits exist on how widely these systems can be deployed.

No True Understanding

AI is pattern matching at a very high level. It has no goals, no desires, no comprehension — it produces outputs that statistically follow from inputs.

Why Aren't Neural Networks Doing Everything?

The Real-world is complex!

It is easy to take billions of digital books and feed it to a computer AI model to learn. It is harder to get that same quality of data from the real-world movements. The more complex the task, the more data you need.

Concerns!

- Ethical concerns of how the data is collected.
- What is the risks of real-world AI interactions?
- (Plugin your favorite Sci-Fi movie in here)
- Who reaps the rewards!?

Solutions?

We will also need different approaches and models to address these issues.

But we also need more data!
And what will that form be?

Possible Benefits!

- In home care?
- Un-bias Policing?
- Dangerous Tasks?
- Loading the dishwasher?
- The list is unending!

Bias in Neural Networks

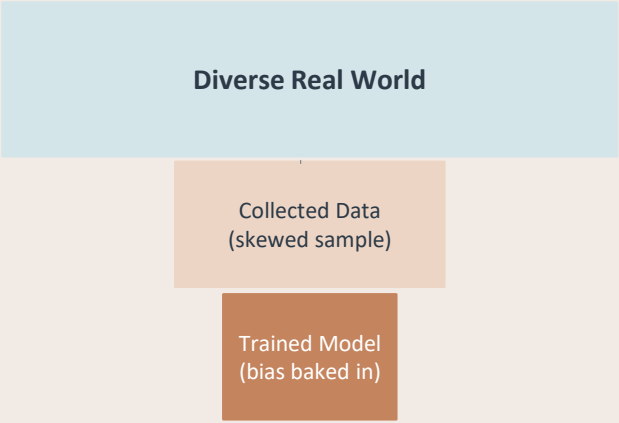
"Garbage in — garbage out"

AI learns from data created by humans — and humans carry biases. If the data used to train a model over-represents certain groups, languages, cultures, or viewpoints, those biases become baked into the model.

This isn't a bug that can be easily patched. It is a **structural feature** of how these systems are built.

Real-world examples:

- Most training data is in English — non-English speakers get worse performance
- Racial and gender biases in facial recognition and hiring tools are well-documented
- Mental health advice may reflect Western, educated, individualistic perspectives
- Rare conditions, minority cultures, and marginalized voices are underrepresented



AI output reflects the bias of training data — not the full human experience

Productivity vs. Learning

We can do more — but are we learning more?

AI dramatically increases what we can produce in a day. A report that once took a week can now be drafted in hours. But there's a question worth sitting with:

If AI did the thinking — did you actually learn anything?

- Expertise develops through struggle — when AI eliminates the struggle, it may eliminate the learning.
- Clinical judgment comes from accumulated experience. Outsourcing thinking to AI may erode that accumulation.
- Many practitioners report a sense of panic when they realize the work is done but they can't explain it or defend it.
- This is especially critical when AI handles complex, nuanced work in human services fields.

The Hidden Costs of AI

Every AI query you send has real-world resource consumption you don't see

~10x

More electricity

per query vs. a regular internet search

500ml

Water per 20-50 queries

for data center cooling systems

1,000s

of data centers

running 24/7 to support AI infrastructure

Growing

Competition for resources

AI data centers compete with homes, hospitals, farms for electricity and water

Young People & Career Anxiety

A generation seeing their dreams disrupted
Social Influencers are the new dream job – because that might be all they see!

Young people — digital natives who rapidly adopted AI — are also among the first to recognize a deeply unsettling reality: the skills and career paths they were building may be significantly disrupted before they even begin.

"My dream job will be AI"	Writers, designers, coders, paralegals, radiologists — young people see AI already doing work they spent years training for
"What's the point of learning?"	If AI can produce a first draft, a logo, or a data analysis in seconds, students question the value of developing those skills
"I don't know how to compete"	Older workers have established expertise; younger workers feel caught between human skills and AI capabilities
Identity and purpose disruption	For many young people, career is tied to identity. AI threat to career can feel like a threat to self

For your practice: this is emerging as a genuine mental health concern. Many young clients will bring AI-related identity and vocational anxiety to sessions.

Cultural Bias & Missing Knowledge

Not everyone's knowledge is being used to train AI — and that has serious consequences

Language gaps

Over 60% of AI training data is in English. Speakers of other languages receive lower-quality, less culturally-relevant responses.

Cultural knowledge gaps

Indigenous healing practices, oral traditions, non-Western therapeutic frameworks, and community-based care models are largely absent from training data.

Geographic inequity

AI is predominantly trained on data from North America, Europe, and East Asia. The lived realities of clients in other regions may be poorly understood by AI.

Intersectional blind spots

Experiences at the intersection of race, gender, disability, class, and culture are underrepresented — exactly the complex situations your clients often present with.

Security

If AI is everywhere then it will be a target for any nefarious actors.

Using sensitive data in LLM

If sensitive data is used in an LLM, it is not automatically incorporated. But a later iteration of the LLM might contain it. Companies that have trained LLMs have used data without asking permission in the past.

Computer System Security

With AI coding and doing quite well, could vulnerabilities be introduced that does not become obvious until too late?

Surveillance

AI cameras and online presence could extend further than ever for companies or countries having an AI profile on everyone.

Bias AI models making bad decisions

Experiences at the intersection of race, gender, disability, class, and culture are underrepresented — exactly the complex situations your clients often present with.

The positive side: Human Brain vs. Neural Networks: The Numbers

	Human Brain	Large AI Model
"Neurons" / Parameters	~86 billion neurons	~1 trillion parameters (GPT-4 est.)
Power consumption	~12 watts	~1,000 watts (inference server)
Efficiency	7 billion neurons per watt	1 billion parameters per watt
Learning from examples	Learns a concept from 1-5 examples	Needs thousands to millions of examples
Energy cost to "learn"	12W running all day, every day	Millions of dollars, months of compute
Generalizes to new tasks	Readily — without retraining	Poorly — often fails outside training domain

PART 4

Using AI for Research

How to ask the right questions — and spot the wrong answers

Step 1: Start with a Clear, Open Question

The Goal: Genuine Curiosity

A good research question is open-ended — it invites exploration rather than confirmation.
You want to understand a topic, not validate what you already believe.

✓ GOOD EXAMPLE

"Why is the sky blue?"

This question is neutral. It doesn't assume an answer. AI can explore all the relevant physics — light scattering, Rayleigh scattering — and give you a comprehensive, balanced explanation.

Key Principle: Your question should invite learning, not confirm what you already think.

Open question = open mind

Step 2: Don't Bake In Your Conclusion

Leading questions lead to biased answers

✗ PROBLEMATIC EXAMPLE

"How does the blue ocean cause the sky to be blue?"

This question contains a false premise — that the ocean causes the sky's color. Because the AI wants to be helpful and the question implies a connection, it may construct a plausible-sounding but incorrect explanation about ocean reflection.

Open Question

"Why is the sky blue?"

AI explores all causes, may mention Rayleigh scattering, light wavelengths, atmosphere. Complete picture.

Leading Question

"How does the ocean make sky blue?"

AI may construct a supportive narrative for a false premise — confidently providing bad information.

Step 3: If You Must Test a Conclusion — Falsify It First

When your research question involves a claim or theory, deliberately ask AI to argue the opposite

1

Your Hypothesis

"Does the blue ocean make the sky blue?"

You suspect a connection between two phenomena

→

2

Ask for the Counter-Argument

"Why does the blue ocean have NO bearing on why the sky is blue?"

Ask AI to explain why your hypothesis might be wrong

→

3

Now You're Informed

Read both sides — then form your conclusion

You understand the strongest arguments for AND against

This technique is called falsification — the cornerstone of scientific thinking. If an idea can't survive being challenged, it probably isn't true.

AI Research — Quick Reference Guide

1. Start open

✓ DO: *"What does research say about grief responses in children?"*

✗ AVOID: *"How does childhood grief cause behavioral problems?"*

2. Remove conclusions

✓ DO: *"What factors influence treatment outcomes for PTSD?"*

✗ AVOID: *"Why does EMDR always produce better PTSD outcomes?"*

3. Ask for the counter

✓ DO: *"What are the strongest arguments AGAINST [theory X]?"*

✗ AVOID: *"Can you confirm that [theory X] is correct?"*

AI Research — Sources

1. Ask for sources

✓ DO: *“Please provide sources! Evaluate if the sources are re"*

2. Investigate

✓ DO: *Make sure the sources are legitimate. Keep in mind most websites are written by AI agents. And Social Media is not a source!*

3. Experts & Data

✓ DO: *If it is an important topic, rely on an expert and make sure they provide data!*

Chiropractors are not doctors!

Remember: AI is a research assistant, not a research authority. Verify important findings with primary sources.

PART 5

AI Resources

Huggingface.co

AI Resources: huggingface.co

How to run things locally and “free”.

How the website is used!

People use Hugging Face and its platform Hugging Face Hub to discover, download, and share machine learning models, datasets, and AI demos. Developers and researchers commonly use it to quickly integrate pre-trained models (like NLP, vision, or speech models) into their applications and collaborate with the AI community.

LLM on your machine: LM Studio

People use LM Studio to run large language models locally on their own computer without needing cloud APIs. Developers and hobbyists use it to download models, chat with them, and integrate them into local apps while keeping data private.

Diffusion: Stability Matrix and DiffusionBee

People use these tools to install, manage, and run different Stable Diffusion tools and models from a single interface on their computer. It helps developers and artists organize AI image-generation packages, update them easily, and generate images locally without complex setup.

Python: Powerful Script Language

Most LLMs can easily code in Python script and even create very capable applications. The reason is that the libraries they use is incredibly powerful and well supported. In a 5 lines of code you can recreate a simplified LLM.

PART 6

Fun Stuff

Huggingface.co

Fun Stuff: LLM

Large Language Models. ChatGPT, Claude, Copilot and Gemini!

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Thank You

Questions & Discussion

Topics we covered today:

- What AI really is — terminology, learning, and costs
- How to research effectively with AI
- Five real concerns about AI in society
- Why the human brain still outperforms AI where it counts